



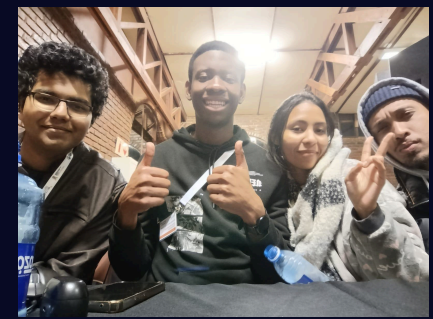
Cloud Computing in HPC: Bare-Metal Provisioning Using Ironic

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Introduction:

This project investigates **how converged cloud technologies can be used to support High-Performance Computing (HPC)**. We explored OpenStack, Kolla Ansible, and Ironic to understand how cloud principles can be applied to HPC environments. Our objective was to deploy and provision a bare-metal cluster using Ironic and compare its performance against a virtualised OpenStack VM-based cluster. The study evaluates scalability, efficiency, and workload suitability across both models.



CSIR Lab Setup

Hardware

- Gigabit switch - connect to a single large managed company switch
- 2x Dell 2U servers in the rack

Networking

- Each server has 4 physical network interfaces
- We assigned static IP addresses to each node within a /24 subnet
- We also used VLAN-tagged interfaces for separation of provisioning, management, and deployment traffic.
- We accessed each node remotely through a WireGuard VPN tunnel using a jumpbox



Figure 1: Network ports 1-4 on server and IDRAC

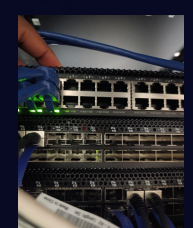


Figure 2: Switch ports 17-20 and 21-24 on switch



Figure 3: 2x Dell 1U servers in the rack

Methods (Provisioning & Deployment):

Ironic

OpenStack Ironic is an open-source component of the OpenStack ecosystem designed to enable bare-metal provisioning.

Bitfrost

Bitfrost is a collection of Ansible playbooks that automates the deployment of base operating systems directly onto physical servers using OpenStack Ironic.

Kolla Ansible

Kolla Ansible is a deployment and management tool specifically designed for OpenStack.

A. Bare Metal Provisioning (Ironic + Bifrost)

This phase involved preparing and provisioning physical compute nodes directly onto hardware using **OpenStack Ironic** and **Bitfrost**. Unlike traditional virtualization, where virtual machines are created, Ironic allows users to deploy and manage physical servers directly, making it ideal for high-performance workloads that demand full hardware access.

After activating the Bifrost environment, the nodes were enrolled into the Ironic service and validated using baremetal node list. Once the nodes reached the "available" state, Ironic deployed the operating system through automated PXE booting. Successful provisioning was confirmed when each node entered the "active" state and allowed SSH access as cloud-user.

B. Deploy the bare metal cluster using OpenStack

After provisioning the hardware nodes, the next step was integrating them into an OpenStack-managed environment. Using Kolla Ansible, the controller and compute services required for cluster operation were deployed in containerized form.

C. Deploy the cluster on VMs using OpenStack

To create a comparable virtualized cluster, OpenStack was used to deploy multiple compute instances (VMs) using the same base operating system image as the bare-metal cluster.

Benchmark Results:

Comparison between Bare Metal Cluster vs Virtual Machine Cluster

- Benchmarking was performed to evaluate and compare the performance of both the bare-metal and VM-based clusters using the stress test (stress-ng).

BM Cluster vs VM Cluster Results		
Metrics	Baremetal Cluster	VM Cluster
Total Operations (bogo-ops secs)	289019	3439530
Real time	30.00	60.06
User Time	239.54	46.28
System Time	0.01	13.78
Throughput (bogo ops/s)	9633.64	1206.53

The benchmark results demonstrate that the bare-metal node delivers significantly superior performance, with approximately 8 times the throughput of the VM cluster. The bare-metal environment exhibits high CPU efficiency with minimal overhead, making it the optimal choice for high-performance computing (HPC) and latency-sensitive workloads. The VM cluster, while offering flexibility, introduces substantial performance penalties due to virtualization overhead.

Comparison between BMC vs VMC		
	Baremetal Cluster	Virtual Machine Cluster
CPU Performance	Excellent - Direct hardware access with no virtualization overhead	Reduced - Impacted by hypervisor overhead and vCPU scheduling
Throughput	High (9,634 ops/sec) - Maximum computational efficiency	Low (1,207 ops/sec) - 8x slower due to virtualization tax
Time distribution	Optimal - 99.99% user time, near-zero system overhead	Inefficient - 30% system time, high kernel overhead
Total Operations	289,019 (in 30 sec) - Higher operations per second rate	3,439,530 (in 60 sec) - Lower ops/second, longer duration required

Challenges & Lessons:

- During deployment, we accidentally edited configuration files inside /etc/kolla_config, assuming they were part of the active setup. Our mentor explained that these files are only used for resets and not the live installation, which caused conflicts. We also learned that changing network interfaces directly in globals.yml can break the entire deployment. The correct method is to override per-node interfaces in /etc/kolla/inventory/overcloud.
- This mistake required a full server reset, we learnt how sensitive Kolla's configuration hierarchy is and how important it is to understand where overrides should be applied.
- Only proceeded with Kolla-Ansible after successful Bifrost provisioning to avoid interface or firmware-related issues.



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Conclusion:

This project gave us real insight into how cloud tools support HPC. By deploying both bare-metal and VM clusters, we saw firsthand how Ironic delivers faster, HPC-focused performance, while VMs offer flexibility for general workloads.

Through setup, troubleshooting, and benchmarking, we gained practical experience with OpenStack and Kolla Ansible, and a clearer understanding of how modern cloud platforms power high-performance systems.